

Executive Summary: This week the team focused on presenting the CDR and starting the steps of building the WAD. After our CDR presentation the team met for a team meeting where hardware members worked on assembling the beginnings of the position module, and software members worked on getting the module actually working. We were able to begin testing with sample data and found that the position module was giving accurate data on Lat/Long position and elevation. The team also put in their last requests for parts for the WAD, and everything has been placed for order but we are still waiting from the ECE office on parts arriving. We should be looking at those arrivals towards the beginning of the next upcoming week. For next week, the team plans on finishing or getting closer to finishing the hardware build of the WAD, such that the software side should be able to begin implementation and testing.

Hardware:

Cody C.

Progress:

- 3/30 - CDR Presentation (~1 Hour)
 - Practiced CDR Presentation with group (~1 Hour)
- 4/1 – Team meeting (~1 Hour 30 Minutes)
 - Talked about next steps for project
 - Updates on parts
 - Waiting on I2S microphone, ordered protoboard and small breadboards
 - Troubleshooted GPS module
- Research I2S configuration (~30 Minutes)
 - Gather information for software testing (Provided code from manufacturer)

Total Hours Worked: ~ 4 Hours

Future Work/Plan

- Wait for I2S Microphone order to come in
- Build Gunshot detection hardware once parts come in

Hardware:

Ryan K.

Progress:

- 3/30 (~2hrs)
 - Practiced CDR slides with group
 - Presented CDR
- 4/1 (~1hr 30min)
 - Acquired more parts that were ordered
 - Built/Plugged-in the GPS for testing
 - Tested GPS accuracy and data collection

Total Hours Worked: (~3hrs 30mins)

Future Work/Plan:

- Obtain the remaining ordered components and fully build the Wearable Alert Device shown from our schematics in preparation for the upcoming testing weeks.

Software:

Alex H.

Progress:

- 3/30 (37 min)

- Practiced CDR slides with team before presentation
- 4/1 (1 h 33 min)
 - Wrote software for GPS functionality.
 - Tested GPS initialization with the team and troubleshooted bugs with the software.
 - Initially, GPS failed to initialize I2C protocol with the microcontroller.
The issue was that the wrong software library (DFRobot_GNSSAndRTC.h) was being used, which resulted in the wrong device address being sent during I2C initialization.
 - Installed the correct software library (DFRobot_GNSS.h) for GPS and rewrote software. I2C initialization became successful.
 - Tested GPS data collection with the team and troubleshooted more bugs.
 - Able to receive accurate timestamp, latitude, longitude, and altitude when outside. However, when inside of a building, the GPS signal is harder to obtain. Only the timestamp is able to be retrieved from the satellite system.
- 4/2 (4 h 33 min)
 - Modified functional GPS software to store timestamp, latitude, longitude, and altitude in data packet C struct.
 - Rewrote wireless data transmission software to send timestamp, latitude, longitude, and altitude.

- Tested wireless data transmission to see if the same data is sent correctly to nRF Connect mobile app. Accurate timestamp, latitude, longitude, and altitude was sent to nRF Connect mobile app wirelessly.
- Tested GPS signal range to determine its current limits. Able to receive GPS signal inside of residential home.
- Computed buffer sizes for 1-second audio buffer and 5-ms audio buffer considering a 44.1 kHz sampling rate and 16-bit sample size (2 bytes). For 1-second audio buffer, 88,200 bytes are needed. For 5-ms audio buffer, 441 bytes are needed.
- Added audio buffers and I2S initialization code for microphone functionality.

Total Hours Worked: 6 h 43 min

Future Work/Plan:

- Continue work on audio data collection software for the microphone.
- Test audio data collection before writing audio signal processing software.

Project Manager:

Jonas W.

Progress:

- 3/30 - CDR Presentation (~1 Hour)
 - Practiced CDR Presentation with group (~1 Hour)
- 3/31 – Finish ordering last parts (~1 Hour)
 - Ordered the microphone, booster board, and the diode.
 - Ordered the small protoboards, and breadboards for building
- 4/1 – Team meeting (~1 Hour 30 Minutes)

- Handed off parts from previous part pickup
- The team has all parts except the booster board, diodes, and microphone.
- Helped with some micro software/hardware implementation of the GPS module.

Total Hours Worked: 4 Hours 30 min

Future Work/Plan:

This upcoming week we plan on getting the last of the parts from amazon and the ECE office.

Once those parts are given I will be sending those off to the team so they can finish building the base of the device. Once the device is finished building I will work with either pushing or getting the testing ordered sooner for our plans of field testing. I will be working on sending out our final presentation slides for the team as well as looking at what it will take to create a holder for the device towards the end of the semester.